

ENTITY ANALYSIS REPORT

QuickBuy – E-Commerce Order Management Database System

1. Introduction

The purpose of this report is to identify the entities, attributes, primary keys, foreign keys, and constraints required for the QuickBuy E-Commerce Order Management Database System.

2. Entity Identification

The following entities are identified in the system:

1. Customer
2. Category
3. Supplier
4. Product
5. Orders
6. Order_Details
7. Payment
8. Shipment
9. Review

3. Entity Attributes and Constraints

Customer Entity

Attribute	Constraint
Customer_ ID	Primary Key, Not Null
Customer Name	Not Null

Attribute	Constraint
Email	Unique, Not Null
Mobile Number	Unique, Not Null
Address	Not Null
Registration Date	Default Current Date

Category Entity

Attribute	Constraint
Category_ ID	Primary Key, Not Null
Category Name	Unique, Not Null
Description	Null Allowed

Supplier Entity

Attribute	Constraint
Supplier_ ID	Primary Key, Not Null
Supplier_ Name	Not Null
Contact_ Number	Unique, Not Null
Email	Unique
Address	Not Null

Product Entity

Attribute	Constraint
Product_ ID	Primary Key, Not Null
Product Name	Not Null

Attribute	Constraint
Price	Check (Price > 0), Not Null
Stock Quantity	Default 0, Check (Stock Quantity >= 0)
Category_ ID	Foreign Key, Not Null
Supplier_ ID	Foreign Key, Not Null

Orders Entity

Attribute	Constraint
Order_ ID	Primary Key, Not Null
Customer_ ID	Foreign Key, Not Null
Order Date	Default Current Date
Total Amount	Check (Total Amount > 0)
Order Status	Default 'Pending'

Order_Details Entity

Attribute	Constraint
Order Detail_ ID	Primary Key, Not Null
Order_ ID	Foreign Key, Not Null
Product_ ID	Foreign Key, Not Null
Quantity	Check (Quantity > 0), Not Null
Unit Price	Check (Unit Price > 0), Not Null

Payment Entity

Attribute	Constraint
Payment_ ID	Primary Key, Not Null
Order_ ID	Foreign Key, Not Null
Payment_ Method	Not Null
Payment_ Date	Default Current Date
Payment_ Status	Default 'Pending'

Shipment Entity

Attribute	Constraint
Shipment_ ID	Primary Key, Not Null
Order_ ID	Foreign Key, Not Null
Delivery_ Address	Not Null
Shipment_ Date	Default Current_ Date
Delivery_ Status	Default 'Processing'

Review Entity

Attribute	Constraint
Review_ ID	Primary Key, Not Null
Customer_ ID	Foreign Key, Not Null
Product_ ID	Foreign Key, Not Null
Rating	Check (Rating BETWEEN 1 AND 5)
Comments	Null Allowed

4. Primary Key Identification

- Customer_ID
- Category_ID
- Supplier_ID
- Product_ID
- Order_ID
- Order_Detail_ID
- Payment_ID
- Shipment_ID
- Review_ID

5. Foreign Key Identification

- Product.Category_ID → Category.Category_ID
- Product.Supplier_ID → Supplier.Supplier_ID
- Orders.Customer_ID → Customer.Customer_ID
- Order_Details.Order_ID → Orders.Order_ID
- Order_Details.Product_ID → Product.Product_ID
- Payment.Order_ID → Orders.Order_ID
- Shipment.Order_ID → Orders.Order_ID
- Review.Customer_ID → Customer.Customer_ID
- Review.Product_ID → Product.Product_ID

6. Conclusion

The identified entities and constraints ensure data integrity, reduce redundancy, and support efficient order management in the QuickBuy E-Commerce platform.